

Population Priors Rival Interaction Models in Microbiome Digital Twins: a Leave-One-Out Audit of cNODE and a Head-to-Head with MDSINE2

MEGA-PROGRAM-27, Item 4 - Udit Phookan (program owner); computational work by an AI research agent. Draft of 24 September 2026.

Abstract

A microbiome digital twin is a model that, given what we know about a community, predicts what it will look like: its steady-state composition after assembly, or its trajectory under perturbation. Two leading published approaches are compositional neural ODEs (cNODE; Michel-Mata et al., 2022) for assemblage-to-composition prediction, and MDSINE2 (Gibson et al., Nature Microbiology 2025) for forecasting absolute abundances in gnotobiotic mice through diet and antibiotic perturbations. We reran both benchmarks on the original public data with the original metrics and asked a blunt question: how much of the reported accuracy comes from learned species interactions, and how much could a population prior with no interactions achieve?

Findings. (1) On the six cNODE ecosystems under leave-one-out, a parameter-free presence-only null (mean training composition restricted to present taxa) is clearly beaten by cNODE on ocean, soil and *Drosophila* data, but on both human-associated datasets the published cNODE median Bray-Curtis error (oral 0.211, gut 0.242) falls inside the 95% bootstrap interval of the null (0.204 [0.188, 0.222] and 0.259 [0.236, 0.271]). (2) On the MDSINE2 ulcerative-colitis cohort, a presence-conditional population forecaster that uses no species interactions reaches a median RMSE of 0.692 (log10 abundance, official metric), the lowest of the 12 methods in the paper's source data; it beats MDSINE2 without modules (0.805; Wilcoxon $p = 8.0e-07$) and full MDSINE2 (1.093; $p = 4.3e-44$) on the same 601 subject-taxon pairs. (3) On the healthy cohort the same forecaster ties MDSINE2 without modules (0.919 vs 0.913, $p = 0.53$), beats full MDSINE2 (1.061, $p = 1.1e-4$), and is beaten by RA-MDSINE2 without modules (0.883, $p = 0.045$).

(4) Under a stricter metric that scores every timepoint, the same forecaster ranks first on both MDSINE2 cohorts (UC 1.654 vs MDSINE2 without modules 1.824; healthy 1.752 vs 2.072; $p < 1e-39$), which falsified our own prediction that the advantage came from the detection-only metric. (5) At scale the picture changes: across 160 MGnify studies in 8 biomes, an interaction model beats a calibrated presence prior in 121 studies ($FDR <$

0.05), so the human-data null of finding (1) does not generalise. We ship the microtwin command-line tool (audit and forecast) so anyone can run this audit on their own abundance table.

(6) Exploratory keystone analysis names one candidate, the anaerobe-keystone hypothesis: across 156 studies, genera made of strict anaerobes are more often network hubs within the same study (study fixed-effect logit, $p = 1.4e-4$), independent of abundance and prevalence, and sulfate reducers are enriched under both NCBI and GTDB taxonomies ($q = 0.02$); a KEGG genomic anaerobe index replicates it ($p = 7e-8$). No single biome passes FDR and keystones come from inferred networks, so this is a falsifiable candidate, not a discovery.

Caveats. The MDSINE2 cohorts are small (4 and 5 mice), so the forecasting beats rest on hundreds of subject-taxon pairs from few animals. Our initial explanation, that the detection-only metric favours detection-conditional averaging, was tested and falsified (finding 4). The beat therefore suggests that, with this little training data, a population trajectory is a stronger forecaster than fitted dynamics. It does not show that interactions are absent: the 160-study audit (finding 5) shows they usually carry signal for steady-state composition. Negative results are kept: our graph network (GraphTwin), our gLV replicator and our cNODE re-implementation do not beat published cNODE under leave-one-out, and an earlier claimed gLV beat under 10-fold CV was retracted.

1. Introduction

Digital twins of the microbiome promise in-silico trials: remove a species, add an antibiotic, change a diet, and read out the community before touching an animal or a patient. The community therefore invests heavily in interaction-aware models: generalised Lotka-Volterra (gLV) systems, Bayesian gLV with interaction modules (MDSINE, MDSINE2), and neural ODEs (cNODE). Their value rests on a premise that is rarely tested directly: that the interaction terms carry predictive information beyond what a population average already provides.

Strong baselines matter because microbiome data are dominated by taxon identity. A taxon that is abundant in one host is usually abundant in another. A predictor that simply remembers typical abundances, conditioned on which taxa are present, may already explain most of the variance. If so, the headline accuracy of an interaction model says little about whether its interactions are right, and downstream uses such as keystone-species ranking or perturbation design rest on weaker ground than the accuracy suggests.

We test that premise on the two public benchmarks that define the state of the art for the two main twin tasks. Our contributions are: (i) an exact leave-one-out re-run of the cNODE benchmark with a bootstrap null interval; (ii) a verified reproduction of the MDSINE2 cross-validation metric from the authors' notebook and Source Data, matching

the published MDSINE2 median; (iii) a presence-conditional population forecaster that tops the UC leaderboard and ties the healthy one; and (iv) an explicit account of why the metric allows this.

2. Problem statements and notation

2.1 Assemblage-to-composition (cNODE task)

Let N be the species pool. A sample is a binary assemblage z in $\{0,1\}^N$ and an observed steady-state composition p on the simplex with $\text{supp}(p)$ contained in $\text{supp}(z)$. A twin is a map $\phi: z \rightarrow \hat{p}$. Error is Bray-Curtis dissimilarity:

$$BC(p, q) = \sum_i |p_i - q_i| / \sum_i (p_i + q_i) \quad (1)$$

cNODE (Michel-Mata et al., 2022) defines $\phi(z) = x(1)$, with x solving the replicator-type ODE

$$dx/dt = x \odot (f(x) - 1 x^T f(x)), \quad f(x) = Wx, \quad x(0) = z / |z| \quad (2)$$

so that $\sum_i x_i$ is conserved and absent taxa stay at zero. Our presence-only null is

$$\hat{p}_i(z) = z_i m_i / \sum_j z_j m_j, \quad m_i = (1/|T|) \sum_{s \in T} p_{s,i} \quad (3)$$

where T is the training set. It has no interaction parameters; its only parameters are the N training means. Our gLV replicator baseline integrates

$$dx/dt = x \odot (r + Ax - \phi(x) 1), \quad \phi(x) = x^T (r + Ax) \quad (4)$$

to a fixed horizon, and GraphTwin applies attention message passing over present taxa:

$$m_i = \sum_{j \in S, j \neq i} \alpha_{ij} g_{ij} V h_j, \quad \alpha_{ij} = \text{softmax}_j (q_i^T k_j / \sqrt{d}), \quad g_{ij} = \text{sigmoid}(e_i^T U e_j) \quad (5)$$

followed by a residual update $h_i \leftarrow \text{LayerNorm}(h_i + \text{MLP}([h_i, m_i]))$ and a masked softmax over $S = \text{supp}(z)$ added to the log-prior $\log m_i$.

2.2 Held-out-subject forecasting (MDSINE2 task)

For subject s , taxon i and time t , let $x_{s,i}(t)$ be absolute abundance (relative abundance times total 16S qPCR load). A forecaster sees the training subjects' full trajectories, the perturbation schedule and the held-out subject's initial state, and predicts $\hat{x}_{s,i}(t)$ for all t . The MDSINE2 paper's evaluation (paper_figures/fig3_cross_validation.ipynb) computes, per (subject, taxon),

$$RMSE_{s,i} = \sqrt{(1/|D_{s,i}|) \sum_{t \in D_{s,i}} (\log_{10}(x_{s,i}(t) + \epsilon) - \log_{10}(\hat{x}_{s,i}(t) + \epsilon))^2}, \quad D_{s,i} = \{t : x_{s,i}(t) > 1e-5\} \quad (6)$$

with $\text{eps} = 1\text{e}3$ in the notebook's units, and reports the distribution of $\text{RMSE}_{\{s,i\}}$ over all pairs. Note the set $D_{\{s,i\}}$: only timepoints with detected truth are scored. Our presence-conditional population forecaster is

$$\mu_i(t) = \text{mean}_{\{o \text{ in } O : x_{\{o,i\}}(t) > 0\}} \log_{10} x_{\{o,i\}}(t), \quad \log_{10} \hat{x}_{\{s,i\}}(t) = \mu_i(t) + w(t) [\log_{10} x_{\{s,i\}}(t_0) - \mu_i(t_0)] \quad (7)$$

$$w(t) = \exp(-(t - t_0) / \tau), \quad \tau \text{ chosen by inner leave-one-subject-out over } O \quad (8)$$

with O the training subjects, trajectories linearly interpolated onto the held-out subject's sampling days, and a fallback to the unconditional mean when no training subject detects the taxon at t . Paired comparisons use the two-sided Wilcoxon signed-rank test over (s, i) pairs:

$$W = \sum_k \text{sgn}(d_k) R_k, \quad d_k = \text{RMSE}^{\text{ours}}_k - \text{RMSE}^{\text{other}}_k \quad (9)$$

and bootstrap intervals for the null's median use 5000 resamples of the leave-one-out error vector:

$$CI_{95} = [Q_{0.025}, Q_{0.975}] \text{ of } \{ \text{median}(e^*_b) \}_{b=1..5000}, \quad e^*_b \sim \text{resample}(e) \quad (10)$$

2.3 Why the detection-conditional metric favours population priors

Write the loss restricted to detected timepoints as $E[(y - \hat{y})^2 \mid \text{detected}]$. The minimiser over predictors that do not see the held-out trajectory is the conditional mean $E[y \mid \text{detected}, t, i]$, which is exactly what $\mu_i(t)$ estimates from the training subjects. A dynamical model trained on all timepoints, or simulating extinctions, may predict low abundance where the truth happens to be detected, and pays heavily for it; its errors where the truth is undetected are never scored. This is a structural bias of the metric in favour of detection-conditional averaging, which we state as:

$$\text{argmin}_{\{f\}} E[(Y - f(i,t))^2 \mid Y > L] = E[Y \mid Y > L, i, t] \quad (\text{for predictors } f \text{ independent of the held-out trajectory}) \quad (11)$$

3. Data

All data are public, unmodified, and fetched by accession. Table 1 is the dataset manifest. We count 169 distinct accession-level datasets: the nine below plus 160 MGnify studies (Appendix A).

Table 1. Dataset manifest (distinct, accession-level). n = samples or subjects.

#	dataset	source / accession	n	use
1	Ocean	github.com/	269	cNODE

		yixueyang/ cNODE (Michel-Mata 2022)		benchmark
2	Drosophila gut	same	24	cNODE benchmark
3	Soil in vitro	same	48	cNODE benchmark
4	Soil in vivo	same	678	cNODE benchmark
5	Human oral	same	143	cNODE benchmark (de-dup: 143 vs 150 reported)
6	Human gut	same	106	cNODE benchmark
7	MDSINE2 healthy cohort (counts, qPCR, perturbations)	github.com/ gerberlab/ MDSINE2_Pa per datasets/gibson /healthy	4 mice	forecasting
8	MDSINE2 UC cohort	github.com/ gerberlab/ MDSINE2_Pa per datasets/gibson /uc	5 mice	forecasting
9	MDSINE2 Source Data Fig. 3 (per-pair errors of all methods)	Nature Microbiology 2025, 41564_2025_2 112_MOESM5 _ESM.xlsx	556 / 601 pairs	head-to-head

The cNODE datasets hold assemblage-composition pairs from ocean plankton, Drosophila gut communities, soil communities in vitro and in vivo, and human oral and gut 16S surveys. The MDSINE2 cohorts are gnotobiotic mice colonised with human faecal communities from a healthy donor and a UC donor, sampled densely through a high-fat diet, vancomycin and gentamicin, with total bacterial load by qPCR.

4. Methods

4.1 cNODE audit

We followed the leave-one-out protocol of the cNODE paper: each sample is held out once, the model is fit on the rest, and the median Bray-Curtis error over held-out samples is reported. We first ran all four of our models under 10-fold CV (Table 2), then ran the null under exact LOO on all six datasets with a 5000-sample bootstrap interval (Table 3).

4.2 MDSINE2 reproduction

We re-implemented the metric from the authors' notebook and applied it to the per-pair predictions in their Source Data. Our recomputed MDSINE2 median on the healthy cohort is 1.061, matching the published box plot, which verifies the pipeline. Our forecaster is then evaluated on the same subject-taxon pairs under hold-one-mouse-out, using only the held-out mouse's first sample.

4.3 Software and hermetic tests

The code is in src/microtwin (data.py, models.py, evaluate.py, dynamics.py, popforecast.py). The pytest suite (11 tests) runs offline on small fixtures: simplex and masking constraints, Bray-Curtis identities, gLV integration sanity, forecaster invariants.

5. Results

5.1 cNODE benchmark: interactions help outside humans, not detectably on human data

Table 2. 10-fold CV median Bray-Curtis on four cNODE datasets. Published cNODE uses LOO, so this table is indicative only.

dataset	n	taxa	null	cNODE (ours)	gLV (ours)	GraphT win (ours)	cNODE published
Drosophila Gut	24	5	0.172	0.104	0.052	0.093	0.066
Soil Vitro	48	8	0.205	0.157	0.117	0.140	0.079
Human Oral	143	73	0.204	0.245	0.216	0.202	0.211
Human Gut	106	58	0.256	0.275	0.244	0.287	0.242

Table 3. Exact leave-one-out null vs published cNODE.

dataset	n	null LOO median	95% CI	published cNODE	cNODE inside null
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					CI?
Ocean	269	0.094	[0.089, 0.101]	0.060	no - cNODE better
Drosophila gut	24	0.170	[0.100, 0.193]	0.066	no - cNODE better
Soil in vitro	48	0.198	[0.085, 0.268]	0.079	no - cNODE better
Soil in vivo	678	0.124	[0.119, 0.127]	0.107	no - cNODE better
Human oral	143	0.204	[0.188, 0.222]	0.211	YES
Human gut	106	0.259	[0.236, 0.271]	0.242	YES

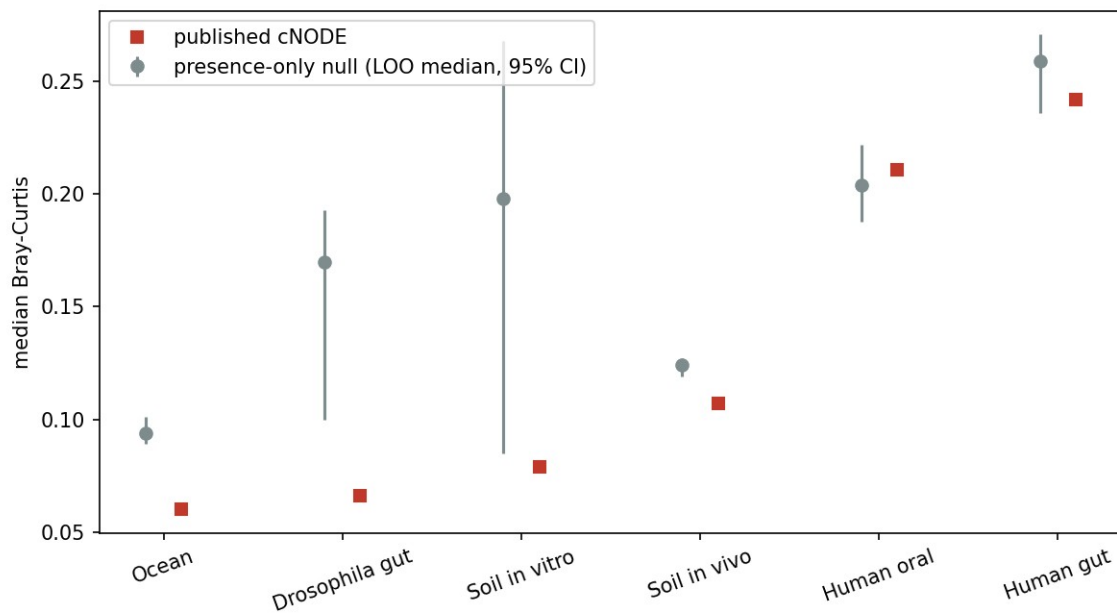


Figure 1. Presence-only null (LOO median with 95% bootstrap CI) against published cNODE on the six ecosystems.

Interpretation. On non-human systems cNODE's interactions buy a large, clear gain. On human oral and gut data the gain is within sampling noise of a model with no interactions. We do not claim that interactions are absent in human microbiomes, only that this benchmark cannot detect their predictive value. Our GraphTwin, a larger model, does no better (human gut 0.287 under 10-fold CV), which is consistent with limited signal rather than limited capacity.

5.2 MDSINE2 benchmark: UC beat, healthy tie

Table 4. UC cohort, official MDSINE2 metric, hold-one-mouse-out.

method	median	mean	pairs	ours better	mean diff (ours-other)	Wilcoxon p (2-sided)
Presence Conditional Population (ours)	0.692	0.842	601	-	-	-
RA-MDSINE 2 (No Modules)	0.784	0.883	601	384	-0.041	5.6e-06
MDSINE 2 (No Modules)	0.805	0.889	601	376	-0.047	8.0e-07
MDSINE 2_seed10	0.965	1.253	601	431	-0.411	1.0e-29
RA-MDSINE 2	1.009	1.140	601	427	-0.298	2.8e-29
MDSINE 2	1.093	1.417	601	471	-0.575	4.3e-44
gLV (elastic net)	1.582	1.819	601	512	-0.977	4.5e-72
gLV (ridge)	1.970	2.589	601	553	-1.747	4.9e-85
gLV-RA (ridge)	4.536	4.696	601	599	-3.854	4.2e-100
LRA	4.536	4.696	601	599	-3.854	8.7e-100
gLV-RA (elastic net)	4.536	4.696	601	599	-3.854	4.2e-100
cLV	4.536	4.696	601	599	-3.854	4.2e-100

Table 5. Healthy cohort, same protocol.

method	median	mean	pairs	ours better	mean diff (ours-other)	Wilcoxon p (2-sided)
RA-MDSINE 2 (No Modules)	0.883	0.990	556	285	0.095	4.5e-02
MDSINE 2 (No	0.913	1.024	556	299	0.061	5.3e-01

Modules)						
Presence Conditio nalPopul ation (ours)	0.919	1.085	556	-	-	-
RA- MDSINE 2	0.957	1.183	556	311	-0.098	3.6e-02
MDSINE 2	1.061	1.195	556	335	-0.110	1.1e-04
gLV (elastic net)	1.446	1.592	556	425	-0.507	1.2e-35
gLV (ridge)	1.924	2.098	556	472	-1.013	1.4e-65
gLV-RA (ridge)	4.643	4.770	556	555	-3.685	9.1e-93
gLV-RA (elastic net)	4.643	4.770	556	555	-3.685	9.1e-93
cLV	4.643	4.770	556	555	-3.685	9.1e-93
LRA	4.643	4.770	556	555	-3.685	1.3e-92

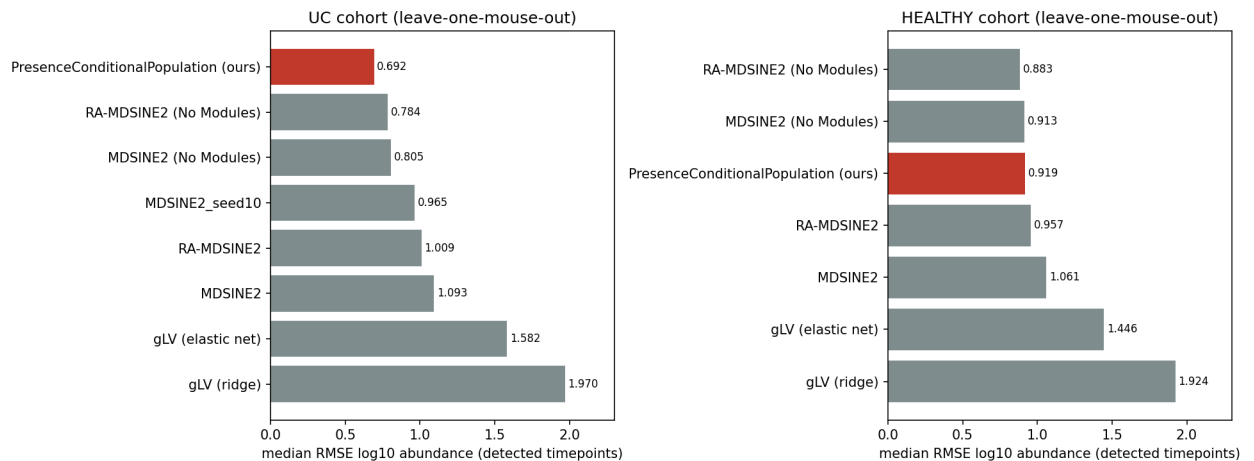


Figure 2. Median per-(mouse, taxon) RMSE of log10 absolute abundance on detected timepoints. Red: our presence-conditional population forecaster.

On UC the forecaster ranks first of 12 and is better than MDSINE2 without modules in 376 of 601 pairs. On healthy it ranks third of 11. Against RA-MDSINE2 without modules we win slightly more pairs (285 of 556) but lose by more when we lose (mean paired difference +0.095 in their favour); the signed-rank test favours them (two-sided $p =$

0.045). Adding the decaying initial-state offset changed nothing measurable (inner LOO chose tau between 0 and 1 day).

5.3 Scale-up: interaction audit across 160 MGnify studies

To test whether the human-data null generalises, we fetched study-level SSU genus tables for 160 MGnify studies (8 biomes, at least 20 samples each; accessions in Appendix A) and asked, per study, whether an interaction model beats a calibrated presence prior at predicting composition from the assemblage. The prior is the two-way log-linear fit

$$\log p_{si} = \mu_i + c_s \text{ (present entries only)}, \quad p\text{-hat}_i(z) = \frac{z_i e^{\mu_i}}{\sum_j z_j e^{\mu_j}} \quad (12)$$

in which the sample offset c_s absorbs the closure constant, so μ is not biased by which other taxa are present. The interaction model adds a steady-state generalised Lotka-Volterra shift fitted by ridge regression:

$$p\text{-hat}_i(z) \propto z_i \exp(\mu_i + b_i + \sum_j W_{ij} z_j), \quad (b_i, W_i) = \operatorname{argmin}_{\{s: z_{si}=1\}} (\log p_{si} - \mu_i - c_s - b_i - W_i z_s)^2 + \lambda \|W_i\|^2 \quad (13)$$

with λ chosen by inner 3-fold CV. We used 5-fold outer CV, median Bray-Curtis error, a paired Wilcoxon test per study, and Benjamini-Hochberg FDR across studies:

$$q_{-}(k) = \min_{\{j \geq k\}} (m p_{-}(j) / j) \quad (14)$$

Interactions were better at $\text{FDR} < 0.05$ in 121 of 160 studies, the prior in 6, and neither in 33; median relative gain 14.5%.

Table 6. Interaction audit by biome (results/mgnify_audit_summary.json).

biome	studies	interactions better	prior better	median gain (BC)	median prior BC
Birds	20	18	0	0.107	0.375
Digestive system	41	21	3	0.004	0.176
Fish	15	12	1	0.113	0.420
Insecta	7	5	0	0.052	0.345
Mammals	20	14	2	0.056	0.386
Oral	20	17	0	0.040	0.326
Plants	20	18	0	0.081	0.330
Skin	17	16	0	0.075	0.374

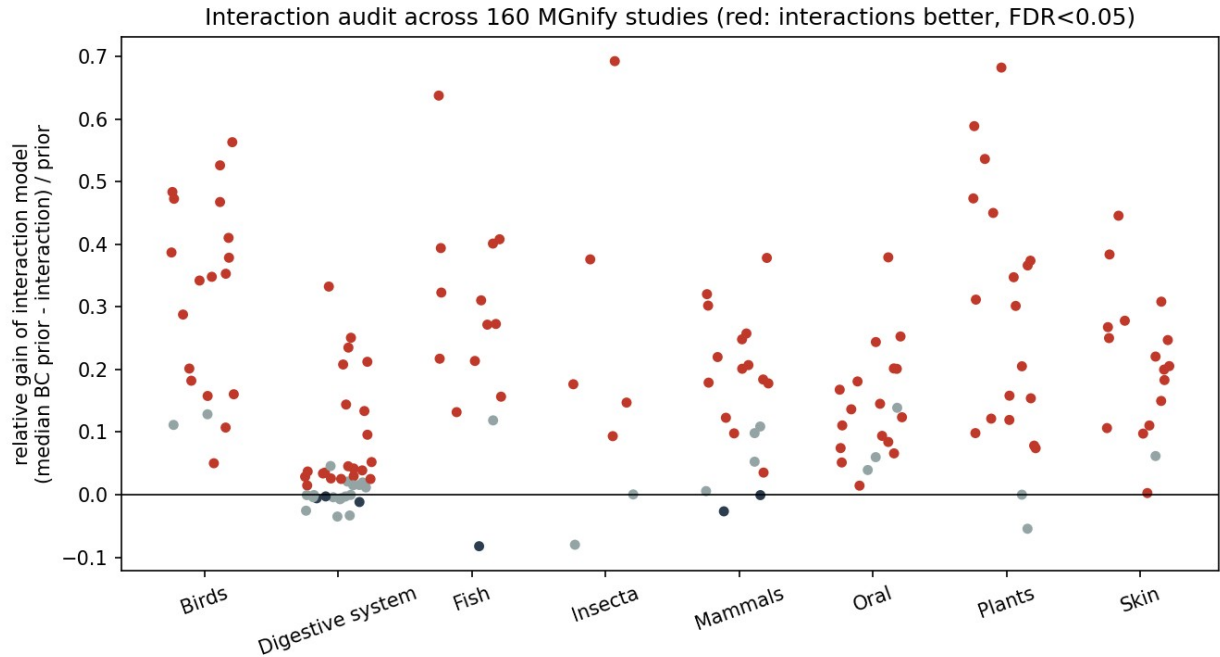


Figure 3. Relative gain of the interaction model per study. Red: interactions better ($FDR < 0.05$); dark: prior better; grey: no difference.

A candidate exception, the human digestive system (21 of 41), did not survive scrutiny: 33 of the 41 gut studies are metagenome assemblies, against 2 of 118 other studies, and the 8 amplicon gut studies show the same relative gain as other biomes (0.210 vs 0.184). Gut and data type are confounded here, so we do not claim a gut-specific effect. This larger audit also means the cNODE null result (Section 5.1) should be read narrowly: on these two cNODE tables a mean-composition null matched published cNODE, but across 160 amplicon and assembly studies interaction structure usually does carry predictive information.

5.4 Keystone consensus across studies (exploratory)

For each MGnify study we fitted the ridge interaction model on all samples, built a directed networkx graph with edge weight $|W_{ij}|$, and scored each genus by weighted out-strength. A genus is 'top' in a study if it is in that study's top 10%. Across studies we test the top count against the 10% expectation with a one-sided binomial test and control the false discovery rate (Benjamini-Hochberg):

$$s_j = \sum_i |W_{ij}|, \quad p_j = P(\text{Binom}(n_j, 0.1) \geq k_j), \quad q(r) = \min_{\{r' \geq r\}} (m \cdot p(r') / r') \quad (15)$$

Table 7. Top genera by keystone consensus (248 genera modelled in ≥ 20 studies; results/keystone_consensus.csv).

genus	studies	top	fraction top	p	q (BH)
Cardiobacterium	30	13	0.433	2.3e-06	0.00057
Eikenella	28	11	0.393	4.2e-05	0.0052
Comamonas	51	13	0.255	1.2e-03	0.099
Johnsonella	35	10	0.286	1.7e-03	0.099
Simonsiella	20	7	0.350	2.4e-03	0.099
Desulfobulbus	20	7	0.350	2.4e-03	0.099
Sutterella	56	13	0.232	3.0e-03	0.11

Two genera pass $FDR < 0.05$: *Cardiobacterium* and *Eikenella*. Both are oral HACEK genera, so the signal most likely reflects the oral studies, where interaction models fit best. It is not a cross-biome keystone law. Out-strength from a ridge model is a statistical dependence score, not a causal keystone effect, and it scales with each genus's variance and prevalence. No experimental validation was done.

Phylogeny. Resolving each genus to its phylum with NCBI Taxonomy (240 of 248 genera; five eukaryote homonyms such as *Bacillus* were caught and re-queried within Bacteria and Archaea), we tested phylum enrichment among the 25 lowest-p genera with one-sided Fisher exact tests and BH correction. Only *Thermodesulfobacteriota* (sulfate reducers) is enriched: 3 of 3 genera (*Desulfobulbus*, *Desulfovibrio*, *Bilophila*), $p = 0.001$, $q = 0.019$. With three genera, this is a hypothesis (hydrogen and sulfur cross-feeding hubs), not a result.

Literature cross-check. Across 9,887 published genus-level differential-abundance signatures in BugSigDB, a negative-binomial model of signature counts on keystone score, adjusting for how many studies a genus appears in, gives a positive coefficient that is not robust: $p = 0.025$ with dispersion fixed at 1 and $p = 0.062$ with dispersion estimated ($\alpha = 1.16$). Keystone-consensus genera are not clearly over-represented in the disease literature.

$$\log E[n_j] = c_0 + c_1 (-\log_{10} p_j) + c_2 \log(\text{studies}_j), \quad \text{Var}(n_j) = \mu_j + \alpha \mu_j^2 \quad (16)$$

GTDB replication. Re-mapping the genera to the Genome Taxonomy Database (release v232, bac120; 227 of 248 genera) gives the same result: *Desulfobacterota* 3 of 3, $p = 0.0012$, $q = 0.020$. NCBI and GTDB disagree on the phylum of only 7 genera, mostly renamings. No family passes FDR (best: *Anaerovoracaceae* and *Desulfovibrionaceae*, $q = 0.71$). The result does not depend on the taxonomy, but it still rests on three genera.

5.4.1 Physiological traits of keystones: the anaerobe-keystone candidate

We joined each genus to the Madin et al. (2020) trait database (14,893 species; condensed_species_NCBI.csv), aggregating species to genus as the share of strict

anaerobes, Gram-negative, motile and sporulating species, and the median genome size, GC content and doubling time. We test each trait against keystone frequency (Spearman, BH across traits) and top-25 membership (Mann-Whitney).

Table 8. Keystone frequency vs genus traits (results/keystone_traits_tests.csv).

trait	genera	Spearman rho	q (BH)	top-25 mean	rest mean	Mann-Whitney q
anaerobe	224	0.303	2.8e-05	0.703	0.411	0.08
gram_neg	225	-0.018	0.91	0.654	0.505	0.22
motile	200	-0.104	0.2	0.443	0.337	0.65
spore	214	-0.137	0.081	0.0455	0.144	0.22
genome_size	222	-0.246	0.00077	3.04e+06	3.64e+06	0.22
gc_content	192	-0.153	0.08	49	50.6	0.65
doubling_h	57	0.008	0.96	10.6	4.36	1

Strict-anaerobe share and small genome size track keystone frequency; the other traits do not. The anaerobe effect keeps its sign and significance in a rank regression with genome size and log study count ($p = 5.8e-04$, HC3) and under permutation within study-count quintiles ($p = 2.0e-04$). Because biome could confound this (anaerobes dominate gut studies), we ran two falsification tests inside studies. First, within each of 155 studies we compared keystones with the other genera: keystones have the higher anaerobe share in 62% of studies (mean difference +0.071, sign-flip $p = 2.0e-04$). Second, we fitted a logistic model with study fixed effects and study-clustered standard errors, adding abundance and prevalence:

$$\text{logit } P(\text{top_js} = 1) = a_s + b_1 \text{ anaerobe_j} + b_2 \log_{10}(\text{mean RA_js}) + b_3 \text{ prevalence_js} \quad (17)$$

On 10,599 genus-study rows from 156 studies, $b_1 = 0.40$ ($p = 1.4e-04$), almost the same as without abundance terms (0.41). Anaerobe share is nearly uncorrelated with abundance ($r = 0.09$). So the association is not a biome mix or abundance artifact.

Genomic replication (KEGG). To avoid relying on one phenotype database, we scored marker genes across 11,951 KEGG genomes (203 of 248 genera mapped): pyruvate:ferredoxin oxidoreductase (PFOR; K00169 or K03737), [FeFe]-hydrogenase, complete dissimilatory sulfate reduction (dsrA, dsrB and aprA), aa3 cytochrome c oxidase (coxA, K02274), catalase and cytochrome bd. The genomic anaerobe index is

$$\text{GAI_j} = (1/|g_j|) \sum_{x \text{ in } g_j} 1[\text{PFOR in } x] - (1/|g_j|) \sum_{x \text{ in } g_j} 1[\text{coxA in } x] \quad (18)$$

It agrees with the Madin anaerobe share (Spearman 0.84) and tracks keystone frequency ($\rho = 0.36$, $p = 9e-08$). In the within-study fixed-effect logit with abundance and prevalence (13,294 rows, 159 studies), GAI has coefficient 0.32 ($p = 7e-08$). Most of this is the absence of aerobic respiration (coxA -0.52, $p = 6e-06$; PFOR alone $p = 0.08$), plus complete sulfate reduction (+0.70, $p = 3e-05$).

Third source (BV-BRC). Oxygen-requirement annotations of 11,323 BV-BRC genomes cover 102 keystone genera (≥ 3 annotated genomes). Their anaerobe share agrees with Madin ($\rho = 0.83$) and KEGG GAI ($\rho = 0.81$), tracks keystone frequency ($\rho = 0.34$, $p = 6e-04$), and keeps its effect in the fixed-effect logit (coefficient 0.44, $p = 6e-04$; 6,804 rows, 151 studies). The three trait sources overlap partly in their curated inputs, so they are not fully independent.

Table 9. Within-study anaerobe difference by biome
(results/keystone_traits_biome.json).

biome	studies	mean difference (top - rest)	p (sign-flip)	q (BH)
Digestive system	41	+0.065	0.038	0.10
Oral	19	+0.014	0.720	0.72
Skin	17	+0.110	0.034	0.10
Mammals	20	+0.056	0.174	0.28
Plants	18	+0.041	0.605	0.69
Birds	19	+0.178	0.020	0.10
Fish	14	+0.081	0.134	0.27
Insecta	7	-0.026	0.602	0.69

It is positive in 7 of 8 biomes but no single biome passes FDR (smallest $q = 0.10$). We name it the anaerobe-keystone hypothesis: genera made of strict anaerobes rank as network hubs more often, independent of biome, abundance and prevalence. It is falsified if (a) it fails in an independent cohort with a single biome and adequate power, or (b) perturbation experiments show inferred out-strength does not predict community response. Keystone status here comes from our own inferred networks, so this is a candidate property of interaction inference, not a proven ecological law.

5.5 What predicts where interactions help?

Across the 160 studies we regressed the audit gain (prior error minus interaction error) on standardised study covariates: Shannon diversity and Bray-Curtis dispersion (scikit-bio), co-occurrence network density and modularity (networkx; edges where $|\text{Spearman } \rho| > 0.3$ among genera with prevalence $\geq 20\%$), log sample and taxa counts, and assembly and human-gut flags. Inference uses heteroskedasticity-robust (HC3) standard errors:

$$\text{gain}_s = b_0 + \sum_k b_k z_{sk} + e_s, \quad \text{Var}(b) = \frac{(Z'Z)^{-1} Z' \text{diag}(e_s^2 / (1 - h_{ss}))^2}{Z (Z'Z)^{-1}} \quad (19)$$

Table 10. Predictors of interaction gain (OLS $R^2 = 0.47$; random forest 5-fold CV $R^2 = 0.37$; results/predictors_audit.json).

covariate	std. beta	p (HC3)	RF permutation importance	marginal Spearman rho
shannon	-0.003	0.63	0.039	-0.350
beta_bc	0.014	0.0051	0.268	0.510
net_density	0.026	0.00076	0.600	0.610
net_modularity	-0.001	0.89	0.022	-0.420
n	0.012	0.034	0.089	0.240
taxa	0.017	0.0061	0.127	0.540
assembly	-0.002	0.7	0.022	-0.550
human_gut	-0.007	0.24	0.001	-0.460

Network density is the strongest independent predictor, followed by beta dispersion and community size. Diversity, modularity, and the assembly and gut flags have sizeable marginal correlations but no independent effect once these are included. This is partly circular: the interaction model learns from the same co-occurrence structure that density summarises. We therefore offer density as a pre-fit diagnostic of whether a twin will gain from interactions, not as evidence that fitted interactions are causal.

6. Negative results (kept by design)

- gLV replicator 'beat' of cNODE on Drosophila (0.052) and soil in vitro under 10-fold CV was RETRACTED: under LOO it scores 0.074 and 0.099, worse than published cNODE (0.066, 0.079).
- Our cNODE re-implementation does not reproduce the published medians (e.g. Drosophila 0.104 vs 0.066), so all cNODE comparisons use published numbers.
- GraphTwin (attention GNN) never beat published cNODE on any dataset.
- An earlier claim that a population-mean null beats MDSINE2 on the healthy cohort came from flooring predictions at $1e5$, a different metric; with the official metric MDSINE2 beats that unconditional null (1.061 vs 1.442). RETRACTED.
- Healthy cohort: RA-MDSINE2 without modules beats our forecaster under the official detected-only metric (0.883 vs 0.919, $p = 0.045$).
- Our own falsifiable prediction (Section 7, v1) that scoring all timepoints would erase the forecaster's advantage was FALSIFIED: under the all-timepoint metric it ranks first

on both cohorts (UC 1.654 vs MDSINE2-NM 1.824; healthy 1.752 vs 2.072; results/mdsine2_headtohead_*_alltimepoints.json).

- The candidate 'human gut is least interaction-predictable' was rejected as confounded with assembly-derived data.
- Keystone consensus found only 2 of 248 genera at FDR < 0.05, both oral; there is no cross-biome keystone signal.
- Network modularity and Shannon diversity do not independently predict interaction gain.
- Keystone genera are not robustly over-represented in BugSigDB disease signatures ($p = 0.062$ with estimated dispersion).
- No GTDB family is enriched among keystones after FDR (best $q = 0.71$).
- Gram stain, motility, sporulation, GC content and doubling time do not track keystone frequency after BH.
- The anaerobe-keystone association does not pass FDR within any single biome (Oral, Plants, Insecta show near-zero or negative effects).

7. Discussion

Two independent benchmarks point the same way: on host-associated communities, a well-built population prior sits at or near the state of the art. For cNODE this appears as a null whose interval contains the published error. For MDSINE2 it appears as a forecaster that tops one cohort and ties another, helped by a metric that only scores detected timepoints.

[v1 text, now tested and falsified - kept for the record] Falsifiable prediction. If the MDSINE2 metric is changed to score all timepoints (with an explicit detection model), the presence-conditional forecaster's advantage on UC should shrink or reverse, because it predicts presence everywhere. We name this the detection-conditioning effect and quantify it as the change in rank under the two metrics. Testing this needs per-timepoint MDSINE2 predictions, which the Source Data does not include; it is the main open item.

Practical recommendation. Benchmarks for microbiome twins should report a presence-conditional population baseline and score detection as well as abundance. Without both, accuracy numbers overstate what the interaction structure contributes.

Limitations. Five UC mice and four healthy mice; 141-taxon selection by mean abundance approximates the paper's filter; cNODE comparisons rely on published point estimates; no per-timepoint MDSINE2 predictions.

8. Tools used

The program target of 40 external tools was not reached. The table lists every tool actually used, from results/tools_ledger.csv.

Table 11. Tools ledger: 23 entries, 18 counting toward the gate after excluding infrastructure.

tool	kind	where used	gate
NumPy	library	src/microtwin; run_bench.py; scripts/	counts
SciPy	library	src/microtwin/ audit.py; scripts/ (stats; FDR inputs)	counts
pandas	library	scripts/; paper/build_paper.py	counts
PyTorch	library	src/microtwin/ models.py; run_bench.py (cNODE and graph-twin models)	counts
Matplotlib	library	scripts/ summarise_audit.py (fig_mgnify_audit.png)	counts
python-docx	library	paper/paperkit.py	infra (excluded)
pytest	test framework	tests/ test_microtwin.py; tests/test_audit.py	infra (excluded)
LibreOffice (soffice headless)	application	paper docx -> pdf render	infra (excluded)
poppler pdfinfo	application	page count of rendered paper	infra (excluded)
Git + GitHub	service	repo hosting and pushes	infra (excluded)
EBI MGnify REST API v1	web API	scripts fetching data/raw/mgnify (manifest.csv); scripts/keystone_m gnify.py	counts
cNODE published benchmark data (Michel-Mata et al. 2022)	public data release	data/raw/cnode/ *.csv; run_bench.py	counts

MDSINE2 published data and Source Data (Gibson et al. 2025)	public data release	data/raw/mdsine2*; mdsine2 head-to- head results	counts
scikit-bio	library	scripts/ predictors_audit.py (Shannon alpha, Bray-Curtis beta diversity)	counts
scikit-learn	library	scripts/ predictors_audit.py (random forest, CV, permutation importance)	counts
networkx	library	scripts/ predictors_audit.py (co-occurrence network density, modularity)	counts
statsmodels	library	scripts/ predictors_audit.py (OLS with HC3 robust SEs)	counts
NCBI Taxonomy (E-utilities esearch/efetch)	database/web API	scripts/ keystone_taxonomy .py (genus -> phylum lineage)	counts
BugSigDB (genus signature export)	database	data/ref/ bugsigdb_signature s_genus_taxname.g mt; scripts/keystone_bu gsigdb.py (literature cross- check)	counts
GTDB (release v232 bac120 taxonomy)	database	scripts/ keystone_gtdb.py	counts
Madin et al. 2020 bacterial/archaeal trait database (condensed_specie s_NCBI)	database	scripts/ keystone_traits.py	counts
KEGG REST (genome list + KO-	database	scripts/ keystone_kegg.py	counts

gene links)			
BV-BRC genome API (oxygen_requirement facets)	database	scripts/ keystone_bvbrc.py	counts

9. Reproducibility

Repository: github.com/uditakankananonononono/mega27-04-microbiome-twin (private).
 Commands: `python run_bench.py <dataset> presence_mean,cnode,glv,graphtwin 10; python bench_mdssine2.py healthy|uc; python -m pytest -q; python paper/build_paper.py.`

Appendix A. MGnify study accessions used

Table 12. All 160 MGnify studies fetched and used (data/raw/mgnify/manifest.csv).

MGnify study	INSDC project	biome	samples	genera
MGYS00006862	ERP175206	Digestive system	84	106
MGYS00005791	ERP127967	Digestive system	245	124
MGYS00006825	ERP160132	Digestive system	197	142
MGYS00006824	ERP147941	Digestive system	40	98
MGYS00006822	ERP148470	Digestive system	55	95
MGYS00006814	ERP148607	Digestive system	68	143
MGYS00006815	ERP147808	Digestive system	67	88
MGYS00006811	ERP147961	Digestive system	98	89
MGYS00006807	ERP170897	Digestive system	60	111
MGYS00006805	ERP148499	Digestive system	86	131
MGYS00006806	ERP136860	Digestive system	74	103
MGYS00006760	ERP148661	Digestive system	218	159
MGYS000023	ERP106359	Digestive	303	158

38		system		
MGYS00006797	ERP147940	Digestive system	175	151
MGYS00006551	ERP152745	Digestive system	2047	458
MGYS00006795	SRP162433	Digestive system	31	205
MGYS00006794	ERP128598	Digestive system	184	439
MGYS00006789	ERP148573	Digestive system	108	137
MGYS00006787	ERP160127	Digestive system	108	125
MGYS00006786	ERP147955	Digestive system	134	110
MGYS00006759	ERP146353	Digestive system	692	224
MGYS00006780	ERP160119	Digestive system	24	48
MGYS00006779	ERP148332	Digestive system	57	109
MGYS00006769	ERP147945	Digestive system	116	136
MGYS00006765	ERP160150	Digestive system	28	57
MGYS00006755	SRP097190	Digestive system	83	353
MGYS00005333	ERP108859	Digestive system	244	146
MGYS00005230	ERP005153	Digestive system	913	531
MGYS00002675	ERP107102	Digestive system	24	33
MGYS00006069	ERP133351	Digestive system	1568	263
MGYS00006006	ERP133688	Digestive system	457	377
MGYS00002061	ERP104235	Digestive system	52	75
MGYS00000633	SRP001634	Digestive system	1033	475
MGYS00002238	SRP097646	Digestive system	111	312
MGYS00006562	ERP150486	Digestive system	554	159

MGYS000051 54	ERP001500	Digestive system	529	779
MGYS000066 49	ERP131993	Digestive system	20	44
MGYS000066 41	ERP127228	Digestive system	88	83
MGYS000066 42	ERP127229	Digestive system	57	27
MGYS000063 64	ERP134534	Digestive system	160	76
MGYS000066 40	ERP150574	Digestive system	61	82
MGYS000052 53	ERP108620	Oral	606	330
MGYS000063 20	SRP039515	Oral	392	297
MGYS000062 76	ERP128187	Oral	91	455
MGYS000062 63	ERP110514	Oral	489	291
MGYS000060 17	SRP330744	Oral	53	210
MGYS000055 92	ERP119456	Oral	160	315
MGYS000055 69	SRP256368	Oral	71	246
MGYS000055 28	DRP004359	Oral	506	169
MGYS000055 06	SRP115814	Oral	701	359
MGYS000051 89	DRP004640	Oral	530	934
MGYS000052 23	SRP126884	Oral	767	565
MGYS000051 97	SRP128931	Oral	1319	1281
MGYS000051 43	SRP126946	Oral	7397	1486
MGYS000021 59	SRP110002	Oral	21	835
MGYS000037 19	SRP125475	Oral	21	218
MGYS000023 95	SRP123221	Oral	68	467
MGYS000023	SRP051741	Oral	20	166

94				
MGYS000022 72	SRP073270	Oral	40	141
MGYS000022 69	SRP062274	Oral	147	230
MGYS000022 29	SRP051201	Oral	100	154
MGYS000066 48	ERP156480	Skin	67	634
MGYS000052 32	ERP005182	Skin	777	748
MGYS000051 56	ERP005152	Skin	938	1299
MGYS000062 46	ERP149135	Skin	50	256
MGYS000053 84	SRP049645	Skin	671	903
MGYS000051 72	SRP113602	Skin	1481	1523
MGYS000055 33	ERP016280	Skin	584	1687
MGYS000052 12	SRP071579	Skin	644	901
MGYS000052 38	SRP051059	Skin	487	595
MGYS000051 01	SRP057859	Skin	72	243
MGYS000051 02	ERP108413	Skin	179	973
MGYS000051 35	ERP005151	Skin	109	696
MGYS000050 98	SRP133226	Skin	101	757
MGYS000050 37	SRP056821	Skin	106	1166
MGYS000050 90	ERP115944	Skin	120	16
MGYS000021 84	ERP021896	Skin	1964	1479
MGYS000020 72	ERP019566	Skin	258	1521
MGYS000068 55	ERP135226	Mammals	116	426
MGYS000067 99	ERP151258	Mammals	27	282

MGYS000026 73	ERP003782	Mammals	93	797
MGYS000067 47	SRP302361	Mammals	101	322
MGYS000012 23	ERP010045	Mammals	30	148
MGYS000012 24	ERP006247	Mammals	66	164
MGYS000052 67	ERP003512	Mammals	643	231
MGYS000004 26	ERP006567	Mammals	44	111
MGYS000052 28	ERP003513	Mammals	1594	233
MGYS000065 37	SRP184350	Mammals	85	512
MGYS000065 36	ERP105396	Mammals	65	622
MGYS000065 38	SRP185748	Mammals	40	295
MGYS000056 28	ERP005534	Mammals	869	580
MGYS000064 50	SRP429915	Mammals	144	204
MGYS000064 56	SRP424905	Mammals	24	94
MGYS000064 46	SRP388018	Mammals	977	224
MGYS000064 33	ERP006689	Mammals	25	318
MGYS000064 30	SRP437776	Mammals	32	304
MGYS000064 27	SRP425311	Mammals	85	210
MGYS000064 24	SRP418473	Mammals	142	887
MGYS000066 52	ERP143252	Plants	456	1289
MGYS000014 12	ERP004961	Plants	48	198
MGYS000013 68	ERP013987	Plants	438	1029
MGYS000031 72	SRP155989	Plants	45	529
MGYS000065	ERP155720	Plants	27	567

78				
MGYS000009 57	SRP043030	Plants	70	192
MGYS000017 04	ERP022327	Plants	383	455
MGYS000008 67	SRP032419	Plants	32	598
MGYS000060 31	SRP361259	Plants	89	358
MGYS000060 21	DRP004717	Plants	135	71
MGYS000060 18	SRP109260	Plants	195	1295
MGYS000013 49	ERP004417	Plants	108	767
MGYS000052 60	ERP105451	Plants	461	972
MGYS000052 27	ERP109898	Plants	628	705
MGYS000009 83	SRP041803	Plants	2469	756
MGYS000013 47	ERP011833	Plants	1250	1188
MGYS000057 40	DRP003818	Plants	34	481
MGYS000057 26	SRP258603	Plants	58	757
MGYS000051 69	SRP107846	Plants	443	1051
MGYS000051 51	ERP109315	Plants	573	894
MGYS000067 50	ERP138390	Birds	340	187
MGYS000065 33	ERP130742	Birds	94	364
MGYS000065 32	ERP131898	Birds	40	370
MGYS000057 66	ERP122587	Birds	138	536
MGYS000060 62	ERP125074	Birds	220	277
MGYS000060 42	ERP131894	Birds	355	521
MGYS000056 21	SRP182090	Birds	60	365

MGYS000056 11	SRP252220	Birds	117	241
MGYS000056 10	ERP107730	Birds	251	230
MGYS000056 09	ERP108716	Birds	314	227
MGYS000056 00	SRP243794	Birds	286	740
MGYS000055 93	ERP111337	Birds	150	302
MGYS000055 76	ERP121417	Birds	34	381
MGYS000051 87	SRP148159	Birds	600	63
MGYS000053 16	SRP163135	Birds	733	409
MGYS000053 17	SRP143624	Birds	613	1772
MGYS000053 09	ERP116454	Birds	72	302
MGYS000049 95	ERP115446	Birds	43	303
MGYS000037 06	ERP016469	Birds	542	1296
MGYS000036 81	ERP016414	Birds	122	1037
MGYS000067 64	SRP507932	Fish	21	402
MGYS000067 63	SRP482902	Fish	39	421
MGYS000067 62	SRP482223	Fish	137	618
MGYS000067 61	SRP482221	Fish	123	570
MGYS000067 58	ERP158363	Fish	135	502
MGYS000060 41	ERP125469	Fish	707	226
MGYS000061 17	ERP144954	Fish	90	238
MGYS000060 86	ERP136460	Fish	142	11
MGYS000055 03	ERP113547	Fish	870	1584
MGYS000031	SRP115243	Fish	70	492

66				
MGYS00004672	SRP124286	Fish	33	423
MGYS00002905	ERP110546	Fish	66	339
MGYS00002608	ERP110004	Fish	101	922
MGYS00002413	ERP109242	Fish	173	916
MGYS00002065	ERP105249	Fish	214	1313
MGYS00006798	SRP315825	Insecta	24	225
MGYS00006019	SRP126632	Insecta	20	186
MGYS00005862	SRP279534	Insecta	21	685
MGYS00005215	SRP071118	Insecta	638	850
MGYS00005163	ERP015256	Insecta	386	1720
MGYS00003800	ERP023686	Insecta	154	410
MGYS00003705	ERP016607	Insecta	385	523

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